

CS & IT ENGINEERING



Data Structure & Programming

Trees and Graphs

Lecture No.- 03



By- Satya sir

Recap of Previous Lecture



- Types of Binary Trees

- FBT, CBT, PBT, SBT, DBT

- Tree Traversal

└ BFT

└ DFT (In order, Pre order, Post order)
(LPR) (PLR) (LRP)

Topics to be Covered



- Practice Q's
- Binary Search Tree
 - Insertion/Construction
 - Deletion
 - Search
- Formulae on Binary Trees.

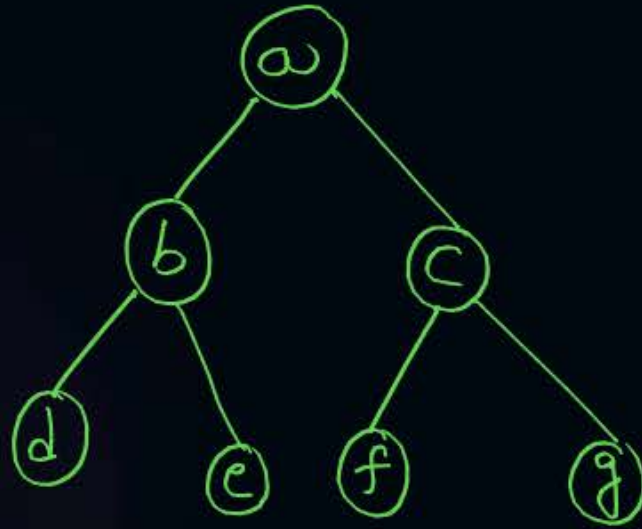


Topic : Binary Search Tree



The in order and preorder traversal of a binary tree are dbefcg and abdecfg, respectively. The post order traversal of the binary tree is:

- (A) ☒ de b f g c a
- (B) e d b g f c a
- (C) e d b f g c a
- (D) d e f g b c a



Post order: d e b f g c a

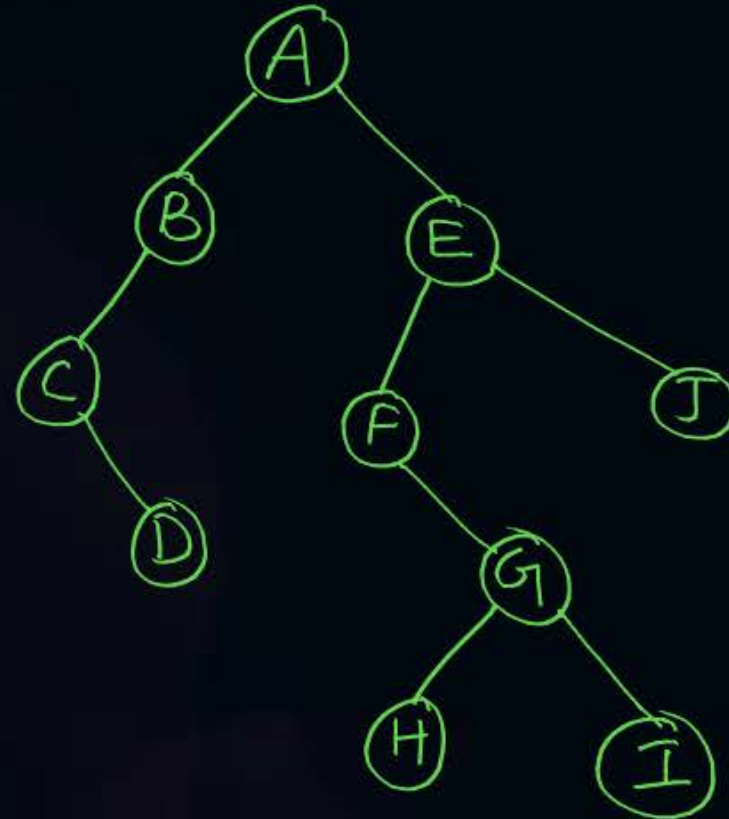


Topic : Binary Search Tree



For a Binary Tree, The In order Traversal Sequence is C D B A F H G I E J and Pre Order Sequence is A B C D E F G H I J. The Post Order Traversal Sequence is _____

- ~~A.~~ D C B H G I F J E A
- ~~B.~~ D C B H I G J F E A
- ☒ C. D C B H I G F J E A
- ~~D.~~ D C B H G F I J E A



Post order

D C B H I G F J E A

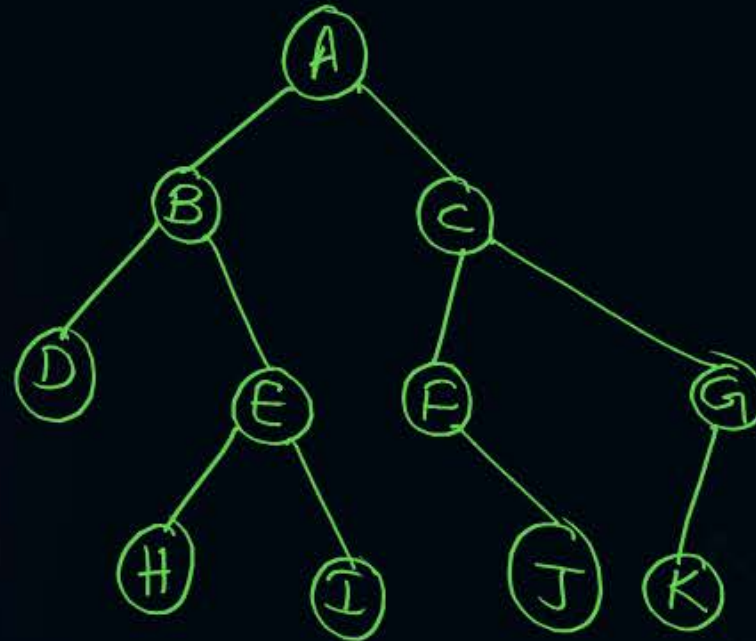


Topic : Binary Search Tree



For a Binary Tree, The In order Traversal Sequence is ~~D~~ ~~B~~ ~~H~~ ~~E~~ ~~I~~ ~~A~~ ~~F~~ ~~J~~ ~~C~~ ~~K~~ ~~G~~ and Post Order Sequence is ~~D~~ ~~H~~ ~~I~~ ~~E~~ ~~B~~ ~~J~~ ~~F~~ ~~K~~ ~~G~~ ~~C~~ ~~A~~. The Pre Order Traversal Sequence is _____

- ~~A. ABDHEICFJGK~~
- ~~B. ABDEHICFJGK~~
- ~~C. ABDEHCIFJGK~~
- ~~D. ABDEHICJFGK~~



Pre order

A B D E H I C F J G K



Topic : Binary Search Tree



Binary Search Tree

A Binary Tree, in which

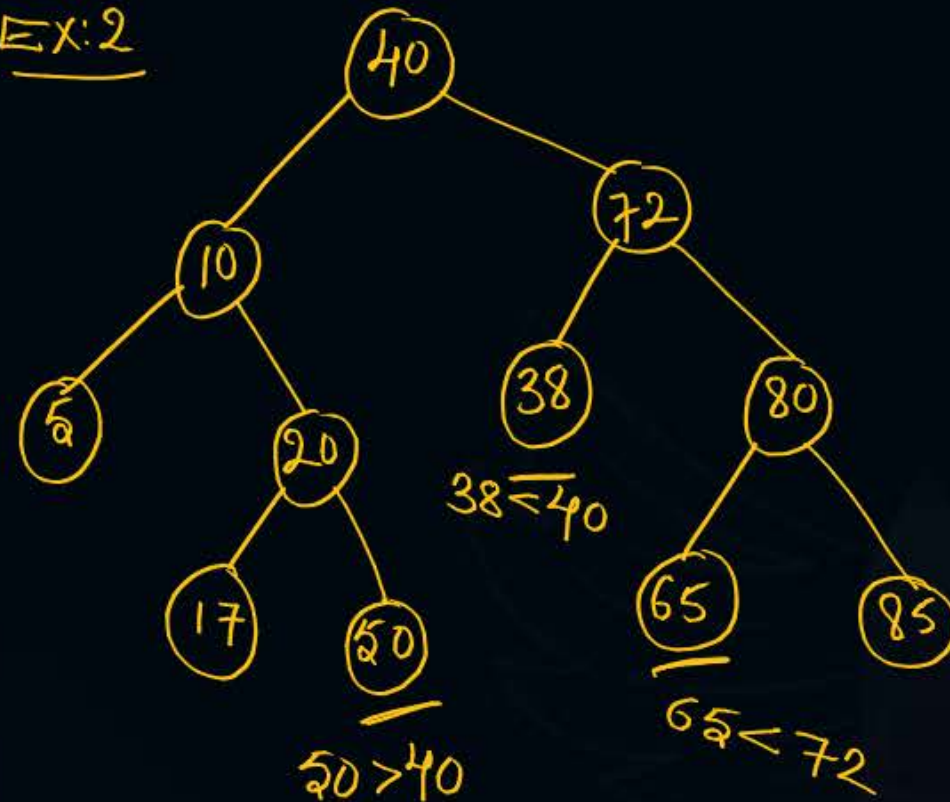
$\text{Left subtree} \leq \text{Parent} \leq \text{Right subtree}$ at Each Node/Level.

Ex: 1



BST

Ex: 2



Not a BST

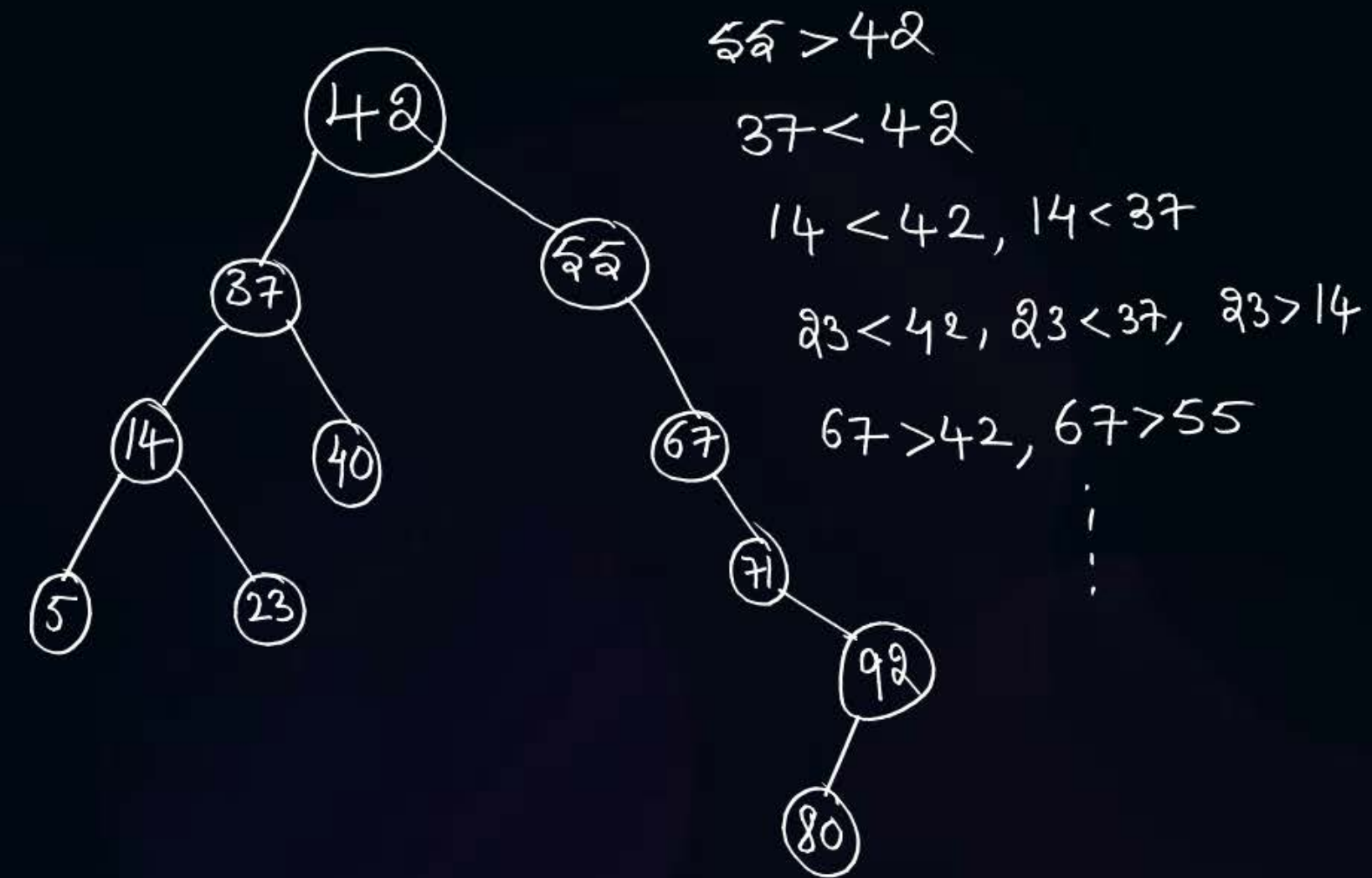


Topic : Binary Search Tree



Insertion / Construction

Construct a BST by Inserting elements in the order : 42, 55, 37, 14, 23, 67, 71, 92, 80, 5, 40



In order Traversal:

5, 14, 23, 37, 40, 42, 55, 67, 71, 80, 92

NOTE: In-order Traversal of BST will always
result in Ascending order of Elements.



Topic : Binary Search Tree

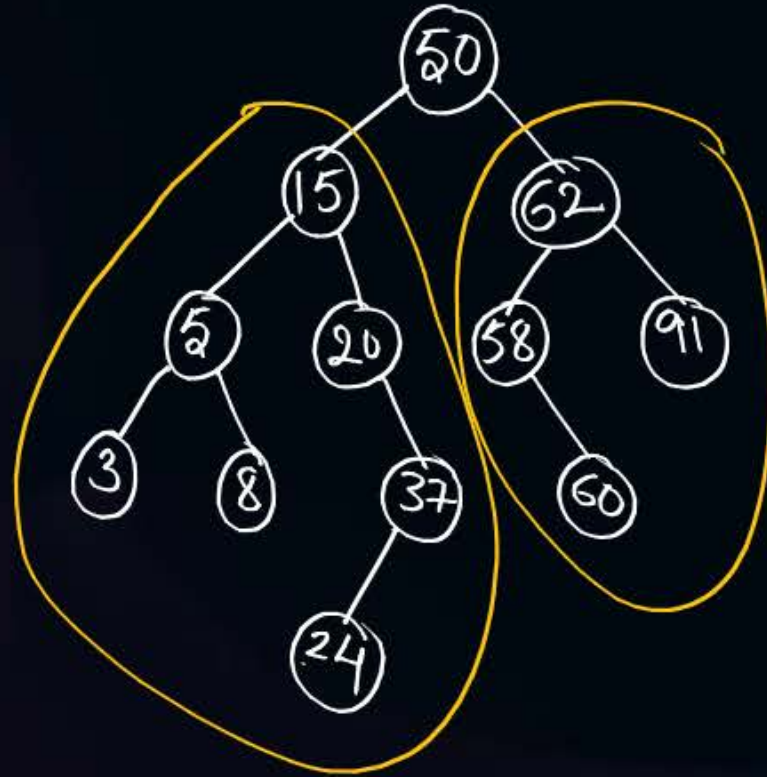


A binary search tree is generated by inserting in order the following integers:

50, 15, 62, 5, 20, 58, 91, 3, 8, 37, 60, 24

The number of nodes in the left subtree and right subtree of the root respectively is

- (a) (4, 7)
- ☒ (b) (7, 4)
- (c) (8, 3)
- (d) (3, 8)



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Post order traversal of a given binary search tree T produces the following sequence of keys

10, 9, 23, 22, 27, 25, 15, 50, 95, 60, 40, 29

Which one of the following sequences of keys can be the result of an inorder traversal of the tree T?

- (A) 9, 10, 15, 22, 23, 25, 27, 29, 40, 50, 60, 95
- (B) 9, 10, 15, 22, 40, 50, 60, 95, 23, 25, 27, 29
- (C) 29, 15, 9, 10, 25, 22, 23, 27, 40, 60, 50, 95
- (D) 95, 50, 60, 40, 27, 23, 22, 25, 10, 9, 15, 29



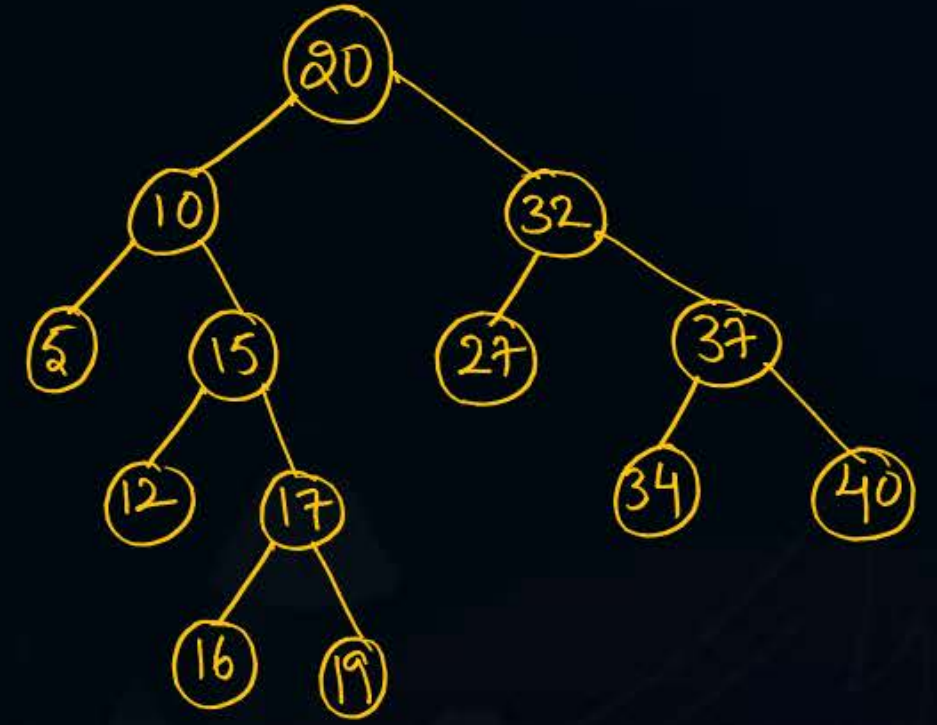
Topic : Binary Search Tree



In: 5, 10, 12, 15, 16, 17, 19, 20, 27, 32, 34, 37, 40

The Post Order traversal sequence for the BST with Pre order sequence as 20, 10, 5, 15, 12, 17, 16, 19, 32, 27, 37, 34, 40 is _____.

- ☒ A. 5, 12, 16, 19, 17, 15, 10, 27, 34, 40, 37, 32, 20
- ☒ B. 5, 12, 16, 19, 17, 15, 10, 34, 27, 40, 37, 32, 20
- ☐ C. 5, 12, 16, 19, 17, 15, 10, 27, 34, 37, 40, 32, 20
- ☒ D. 5, 12, 16, 19, 17, 15, 10, 27, 37, 40, 34, 32, 20



5, 12, 16, 19, 17, 15, 10, 27, 34, 40, 37, 32, 20

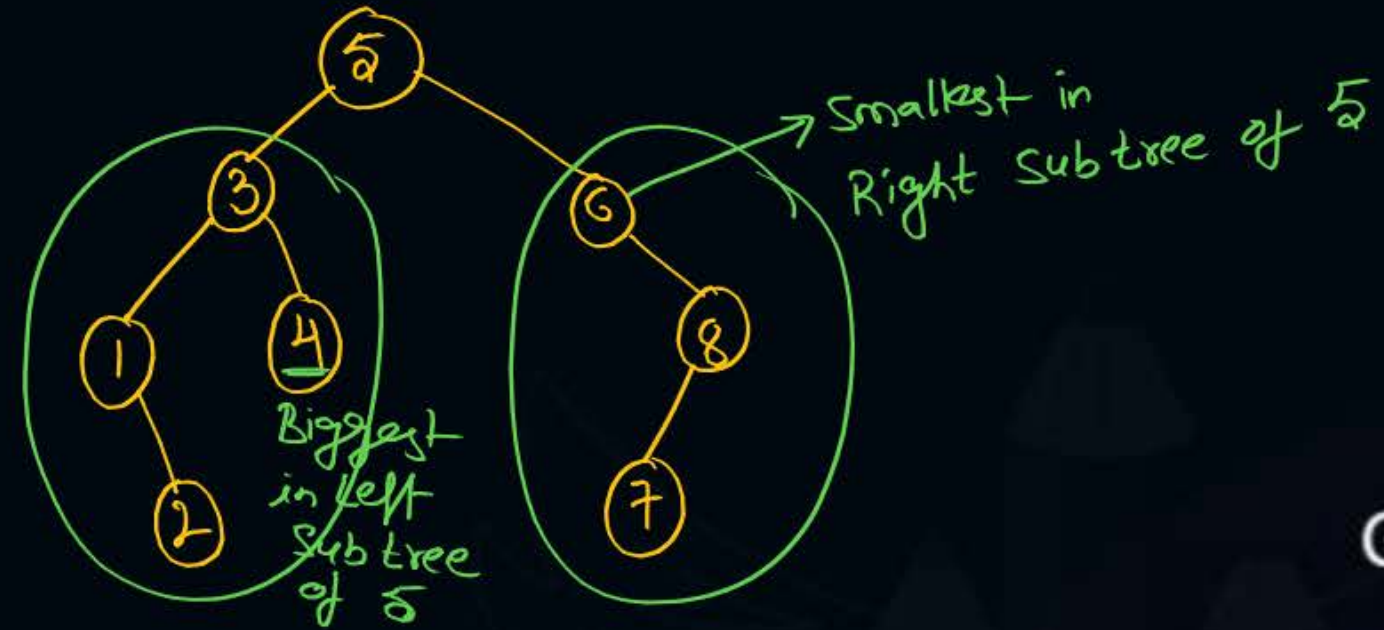


Topic : Binary Search Tree



A binary search tree contains the numbers 1, 2, 3, 4, 5, 6, 7, 8. When the tree is traversed in pre-order and the values in each node printed out, the sequence of values obtained is 5, 3, 1, 2, 4, 6, 8, 7. If the tree is traversed in post-order, the sequence obtained would be

- (A) 8, 7, 6, 5, 4, 3, 2, 1
- (B) 1, 2, 3, 4, 8, 7, 6, 5
- (C) 2, 1, 4, 3, 6, 7, 8, 5
- (D) 2, 1, 4, 3, 7, 8, 6, 5



Post-order : 2, 1, 4, 3, 7, 8, 6, 5

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Topic : Binary Search Tree



In order Predecessor : Biggest Element in Left Subtree

In order Successor : Smallest Element in Right Subtree.

Deletion of a Node from BST

Case-1 : Delete a Node with Zero child (Leaf node) : Simply Delete it.

Case-2 : Delete a Node with One child

- Left child : Swap Node with In order Predecessor, Repeat it until Node becomes leaf node. Then Delete it.
- Right child : Swap Node with In order Successor, Repeat it until Node becomes leaf node, Then Delete it.

Case-3 : Delete a Node with Two children

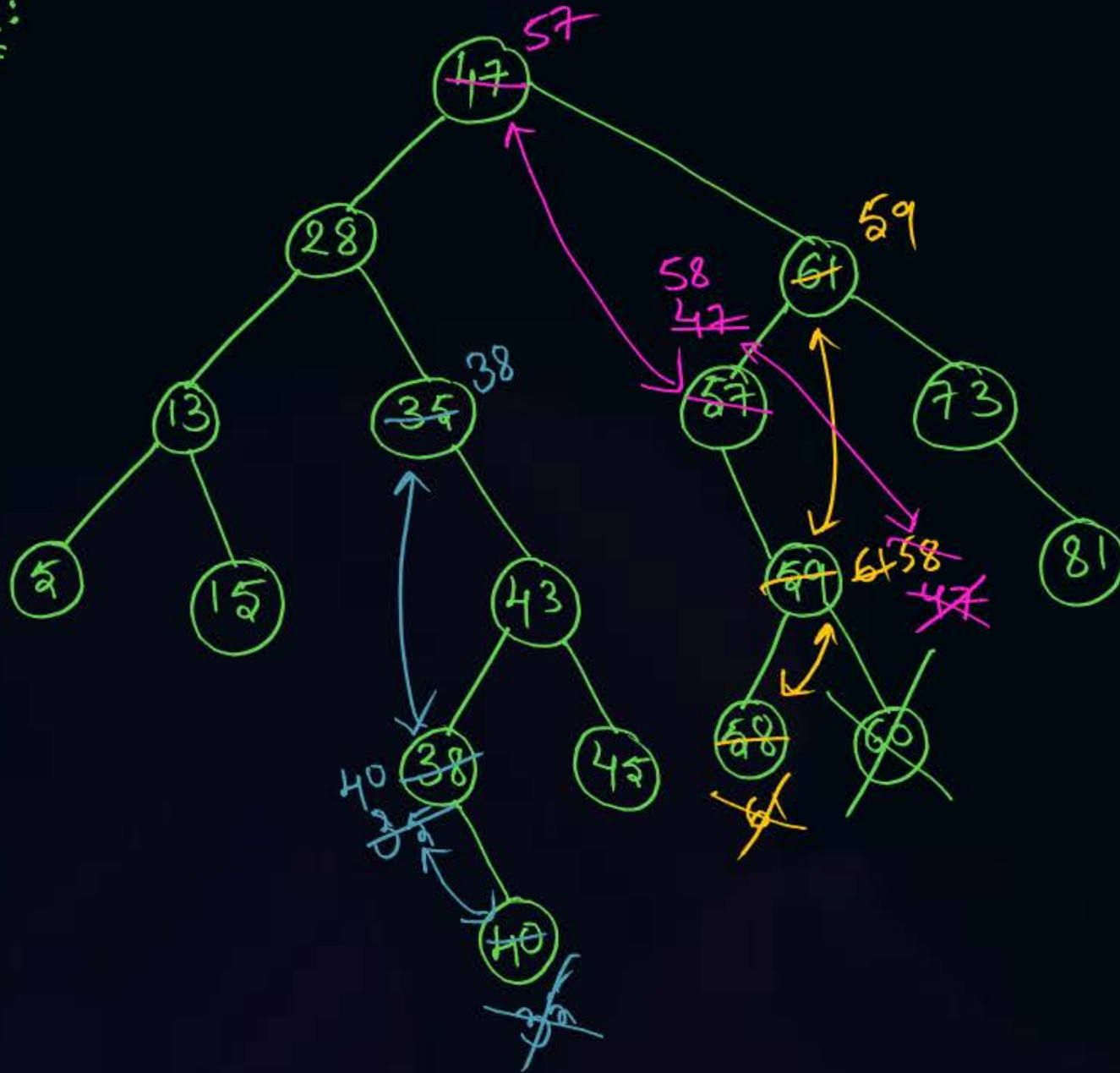
- ↓ (default)
- Swap with In order Predecessor
- Repeat until it reach leaf
- Delete it
- Swap with In order Successor
- Repeat until Node becomes Leaf
- Delete it.



Topic : Binary Search Tree



Ex:



Delete 60 — Simply Delete it, free up memory.

Delete 35 — Swap with 38, Then Swap with 40; Delete 35.

Delete 61 — Swap with In order Predecessor (or) Successor.

— Swap 61 with 59

— Swap with 58

— Delete 61

Delete 47: Swap with In order Predecessor (or) Successor.

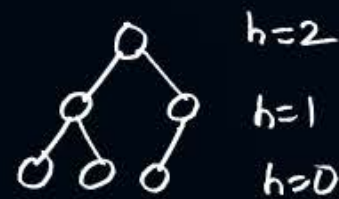
— Swap 47 with 57

— Swap with 58

— Delete 47



Height = 5



$$\left\lceil \log_2^{6+1} \right\rceil - 1 = 3 - 1 = 2$$

- ⑤ Given 'N' Nodes, The Maximum height Possible = $(N-1)$ [SBT/DBT]
The minimum height Possible = $\lceil \log_2^{N+1} \rceil - 1$ [PBT/CBT]

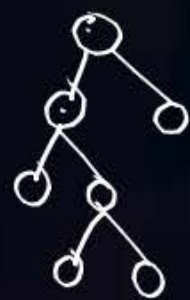
- (6) Given Height 'H'. The minimum nodes required to form a Binary Tree with height H $\Rightarrow$ H+1 [SBT/DBT]
The Maximum nodes " " " " " $\Rightarrow 2^{(H+1)} - 1$ [PBT]

- ⑦ In a Binary Tree, if there are 'i' leaf nodes Then, No. of nodes with exactly 2 children = i-1



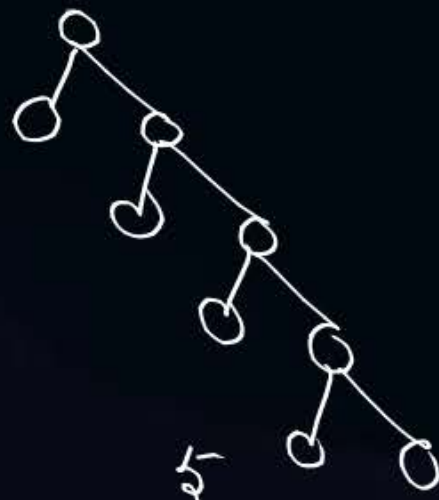
$$g = 2$$

No. of nodes with $= 2-1$
Exactly 2 children $= 1$



4

3



5

4

- ⑧ In a Binary Tree, if there are 'N' Nodes,
Then No. of Edges will be $N-1$.



How many distinct binary search trees can be created out of 4 distinct keys?

- (A) 5
- (B) 14
- (C) 24
- (D) 42

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2 mins Summary



- BST
 - Construction
 - Insertion
 - Deletion
- Formulae on Binary Trees

To be contd...





THANK - YOU